



A supernova is a massive exploding star that shines very brightly, giving the impression that a new star has appeared. Tycho Brahe first described the phenomenon in 1573.

number of editions, eventually becoming a 20-volume work. In one of the later editions of around 1570, he included a description of the **camera obscura**. The principles of this device date back to China and Greece around 2,000 years before, but della Porta was the **first to suggest the use of a convex lens** rather than a pinhole to focus the image on the screen inside, allowing more light through the larger aperture without a subsequent loss of clarity. He also used this camera obscura with a lens, an innovation that was of great value to Kepler in the following century in his studies of the workings of the human eye (see 1598–1604).

Although a successful attorney in Paris, **François Viète** (1540–1603), also known as Franciscus Vieta, was a talented mathematician who devoted much of his spare time to the subject. One of his first achievements in the field was a set of **trigonometric tables** to aid calculation, the *Universalium Inspectionum ad Canonem Mathematicum Liber Singularis*, which he started to publish in 1571.

COMPOSING A BOOK ON ALGEBRA that was both comprehensive and intelligible to the nonmathematician, **Rafael Bombelli** (1526–72) published his treatise simply titled *Algebra* shortly before his death in 1572. He explained in everyday language the algebra known at that time and tackled a problem barely understood by his contemporaries, **imaginary numbers**: numbers whose **square is less than zero**.

These, he explained, cannot be dealt with in the same way as other numbers, but are essential in solving equations involving powers of two, three, or four. Bombelli effectively laid down the rules for using these imaginary numbers for the first time. Despite his pioneering work, imaginary numbers were not accepted in mathematics until almost 200 years later.

Tycho Brahe had observed what appeared to be a bright new star in the **constellation of Cassiopeia** in 1572, and the following year published his account

Astronomical clock

This astronomical clock, built between 1547 and 1574, stood in Strasbourg Cathedral until the 19th century. An 1840s replica now exists in its place.

“IT WAS NOT JUST THE **CHURCH THAT RESISTED THE HELIOCENTRISM OF COPERNICUS.**”

Tycho Brahe, Danish astronomer, 1587

of it, *De Nova Stella* (*On the New Star*). It was in fact a **massive stellar explosion**, not a new star, but the Latin word “nova” later came to be applied to what is now known as a **supernova**,

as well as to other stars that brighten abruptly. Brahe realized that his “star” was very distant from Earth, certainly beyond the orbit of the Moon. Conventional wisdom since the time of Aristotle had maintained that anything outside the immediate vicinity of Earth was unchanging, including the stars of the celestial realm, but Brahe’s observations **contradicted the idea of the immutability of the stars**.

One of the most elaborate **astronomical clocks** of the period was built in the Cathedral of Notre Dame in Strasbourg to replace the existing 200-year-old clock, which had stopped working. The new clock was designed by mathematician **Christian Herlin** in the 1540s, but only the preliminary building had been accomplished in 1547 when work was interrupted by Herlin’s death. Political problems further delayed work until the 1570s, when Herlin’s pupil, mathematician



TYCHO BRAHE
(1546–1601)

Born in Scania, now in Sweden but then part of Denmark, Tycho Brahe became interested in astronomy while studying law in Copenhagen. Under the patronage of King Frederick II of Denmark he established an observatory equipped with the finest astronomical instruments.

Conrad Dasypodius (1532–1600), took over the project. The clock was finally built by Isaac and Josias Habrecht. It incorporated many of the latest ideas in mathematics and astronomy—as well as clockmaking—into its design, which included a celestial globe, an astrolabe, a calendar dial, and automata.



animated figures

staircase

1571 François Viète begins publication of trigonometric tables

1572 Rafael Bombelli publishes Algebra

1573 Danish astronomer Tycho Brahe publishes De Nova Stella

1574 The astronomical clock in Strasbourg Cathedral is completed